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Bridging urban and individual sustainability: how urbanisation shapes creativity via brain structure

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Understanding the interplay among urbanisation, innovation, and individual creativity is essential for informing effective policies and promoting sustainable development that benefits both individuals and society. Using behavioural and brain data of 3480 participants (2012–2023) in China, along with urbanisation and innovation indexes across provinces (2007–2022), we conducted coupling coordination and multilevel analyses to investigate these relationships and the role of brain structural plasticity. Results demonstrated that individual creativity was positively associated with both urbanisation and innovation, and provinces with a higher level of urbanisation exhibited more closely coupling coordinated development among the three indices over time. Notably, the mean urbanisation index of the last 5 years can significantly predict individual creativity. Mediation analyses further indicated that variations in brain volume within the default mode, somatomotor, and central executive networks significantly impacted the effect of urbanisation on shaping individual creativity. These findings first establish a direct link among urbanisation, innovation, individual creativity, and neuroanatomical changes, offering new insights and empirical evidence of the mechanisms bridging macro-level urban sustainability with micro-level individual sustainability.

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Introduction

Urbanisation is a complex socio-economic process that involves the expansion of urban areas (Liu et al., 2020) and the migration of rural populations (Zhang et al., 2022), leading to increased population density, information flow, trade, and transportation. This process drives significant shifts across multiple dimensions of social life, including occupations (e.g., the transition from agricultural labour to industrial and service sectors), lifestyles (e.g., divergent living patterns arising from disparities in access to urban infrastructure and services), cultures (e.g., increased cultural diversity), as well as human behaviour and well-being (e.g., changes in marital and reproductive decisions, access to healthcare, and exposure to heightened environmental risks) (Montgomery et al., 2003; Gong et al., 2012; United Nations Department of Economic Social Affairs, 2019; Tu et al., 2025). By concentrating technology-related resources, institutions, and skilled human capital, urbanisation promotes the generation and diffusion of innovation (Storper and Scott, 2009; Glaeser, 2012). In turn, urban innovation helps address challenges during urbanisation process, such as climate change, resource depletion, and environmental pollution, by advancing technology and institutional development (Hart and Milstein, 2003; Schumpeter, 2013). This close nexus between urbanisation and urban innovation largely promotes sustainable development of society, economy, and environment (Nakao and de Andrade Guerra, 2020). Following growing recognition that sustainability processes operate across multiple social scales (Saleh and Brem, 2023), we extend this perspective to the individual level by conceptualising individual sustainability as individuals' capacity to adapt effectively to dynamic urban and social environments, including the development of human capital capacities that support innovation and long-term adaptation.

Individual creativity—an essential component of human capital—is increasingly recognised as a key competency for sustainable development, as it underpins innovation and enables the generation of novel and useful solutions to complex sustainability challenges, thereby supporting sustainability transitions (Malik, 2018; Saleh and Brem, 2023). In this context, we view individual creativity as a key behavioural manifestation of individual sustainability. Importantly, as a socially embedded capability, individual creativity is closely linked to both urbanisation and innovation (Udwadia, 1990; Bloom et al., 2008). According to the componential theory of creativity (Amabile, 1983, 1988, 1996, 2011), individual creativity emerges from the dynamic interaction between personal traits and the external environment, including social and cultural contexts. Consistent with this theoretical framework, empirical evidence suggests that migration from rural to urban areas can enhance children's creative thinking and motivation as it exposes them to heterogeneous environments that stimulate novel ideas and enhance personality openness, challenges them to adapt to new cultural norms that break habitual thinking, and provides them with educational resources that support creative exploration (Shi et al., 2012, 2013). In a broader sociocultural context, urbanisation shapes the environment by increasing cultural diversity (Fischer, 1975; Knox and Pinch, 2014), developing intellectual ecosystems (Black and Henderson, 1999; Bertinelli and Black, 2004; OECD and Commission, 2020; Chan and Chan, 2022), cultivating the atmosphere of innovation (Pred, 1966; Higgs, 1971; J. Zhang et al., 2023), and influencing culture tightness——referring to the strength of social norms and the degree of sanctioning for deviance——which shows a context-dependent association with urbanisation, tending toward looseness globally but higher tightness in highly urbanised regions of China, with implications for innovation and creativity (Gelfand et al., 2011; Harrington and Gelfand, 2014; Chua et al., 2019). These factors may facilitate the translation of creative potential into observable creative achievement in real-world contexts. From this perspective, broader urban creativity discourses and policies manifest through the creative engagement of individuals embedded within these environments; thus, individual creative achievement serves as a crucial behavioural indicator reflecting how urban contexts facilitate the translation of creative potential into socially recognised outcomes. At the same time, the promotion of creative urbanisation yields ambivalent effects (Borén et al., 2013). Specifically, the effects of urbanisation on creativity and innovation may be heterogeneous across social contexts, as urban environments can simultaneously expand opportunities for knowledge exposure, innovation participation, and creative resource accumulation, while also generating regional inequalities and uneven access to creative opportunities (Florida, 2003; Peck, 2005). Despite these advances, the direct associations between urbanisation, innovation, and individual creativity——and the potential mechanisms through which urbanisation may shape creativity——remain largely underexplored.

One potential mechanism through which urbanisation and innovation influence individual creativity is cumulative exposure. Previous studies suggest that environmental effects on individual behaviour and brain phenotypes often emerge gradually, as their impact accumulates over time rather than manifesting immediately (Pope III et al., 2009; Grandjean and Landrigan, 2014; Lacagnina, 2020; Xu et al., 2022). The 4C Model of Creativity (Kaufman and Beghetto, 2009) also highlights that the development of individual creativity relies on the continuous integration of novel experiences, learning opportunities, and exposure to cultural and social contexts. The accumulation of internal experiences and external resources over time is essential for shaping individual creative abilities (Amabile, 1996; Fogarty et al., 2015). Based on these factors, we hypothesised that urbanisation may shape individual creativity by gradually accumulating supportive resources, resulting in temporally lagged effects. In this study, we examine the relationships between urbanisation, innovation, and individual creativity, with a particular focus on testing the hypothesis of lagged and cumulative effects of urbanisation on individual creativity.

Urbanisation not only shapes individual creativity through socio-economic factors but may also influence its neurobiological foundation. Neuroimaging studies have demonstrated that individual creativity is closely associated with brain structural and functional phenotypes (Jung et al., 2010; Chen et al., 2014; Bashwiner et al., 2016; Zhu et al., 2016; Shi et al., 2017; Wertz et al., 2020). For instance, creative achievement has been associated with the neuroanatomical variations in the precentral gyrus (Zhu et al., 2016), precuneus cortex (Chen et al., 2014; Wertz et al., 2020), and orbitofrontal cortex (Jung et al., 2010). Artistic creativity has been linked to the surface area of the orbitofrontal cortex (Bashwiner et al., 2016) and paracentral gyrus (Shi et al., 2017), as well as the grey matter volume of the precentral gyrus (Anic et al., 2018). Cognitive flexibility, a core component of creativity, has been correlated with the functional connectivity within the prefrontal cortex (Chen et al., 2014). Moreover, prolonged exposure to urban environments has been shown to enhance the development of brain regions involved in higher-order cognition, particularly the prefrontal cortex (Xu et al., 2022; X. Zhang et al., 2023). In contrast, adverse urban exposures hinder the neural development of various brain regions, including orbitofrontal cortex (Beyer et al., 2024), dorsolateral prefrontal cortex (Meyer-Lindenberg, 2010; Van Os et al., 2010; Haddad et al., 2015; Lambert et al., 2015; Xu et al., 2023; Beyer et al., 2024), precentral gyrus (Xu et al., 2023; Beyer et al., 2024), and supplementary motor area (Xu et al., 2023), potentially

influencing psychological and behavioural outcomes (McMichael, 2000; Meyer-Lindenberg, 2010; Van Os et al., 2010; Lederbogen et al., 2011; Haddad et al., 2015; Lambert et al., 2015; Schumann et al., 2023; Xu et al., 2023; Beyer et al., 2024; Polemiti et al., 2024). These findings led us to speculate that brain structural phenotypes may mediate the relationship between urbanisation and individual creativity.

To investigate the relationship among urbanisation, innovation, and individual creativity, we collected behavioural and brain data of 3480 participants (2012–2023), alongside provincial-level urbanisation and innovation indexes (2007–2022) within the context of China's "new-type urbanisation"——a people-centred and quality-oriented strategy first proposed at the 18th CPC National Congress in 2012 (Hu, 2012), which emphasises innovation-driven, coordinated, green, open, and inclusive development, and focuses on urban–rural integration, equalisation of public services, modernisation of governance capacity, ecological sustainability, and cultural heritage preservation, as articulated in the *National New-type Urbanisation Plan (2014–2020)* (State Council of the People's Republic of China, 2014). Utilising this comprehensive dataset, we first examined the associations and coupling coordination among urbanisation, innovation, and individual creativity. We then conducted lagged analyses to assess the cumulative effects of urbanisation on individual creativity over time. Finally, integrating neuroimaging data, we performed mediation analyses to examine whether the brain neuroanatomical variations mediated the effects of urbanisation on individual creativity, while controlling for personality traits, intelligence, and genetic factors.

Methods

Participants. This study utilised behavioural, neuroimaging, and genetic data from four healthy adult cohorts (total $N = 3503$) of the Southwest University Longitudinal Imaging Multimodal project (SLIM), Gene–Brain–Behaviour Project (GBB), longitudinal Gene–Brain–Behaviour Project (LGBB) and Behavioural Brain Research Project of Chinese Personality (GGBBP), collected at the Brain Imaging Center of Southwest University from 2012 to 2020. In addition, a supplementary behavioural dataset ($N = 884$, including residential information and creativity measurements) was also included in this study. After removing outliers and missing data, the behavioural analysis consisted of 3480 participants (mean age = 19.61 ± 1.76, 33% male, 67% female). Among them, 2606 participants had structural T1-weighted MRI data, and 1585 participants had both structural MRI and genetic data (Fig. 1). It is important to note that our genetic data come from two sources. The first source includes genome-wide data for 1044 participants from previously published studies (Chen et al., 2020; Feng et al., 2020, 2022; Liu et al., 2023; Wang et al., 2024), specifically from the SLIM ($N = 286$), GBB ($N = 433$), and LGBB ($N = 325$) datasets. After matching with brain and creativity achievement data, the available data for these datasets included 206, 363, and 258 individuals, respectively. The second source is the GGBBP dataset, which originally contained genome-wide data for 935 individuals; after matching, 758 individuals were included in the final analysis. Further detailed information on each dataset is available in the Dataset information section of Supplementary Notes. All participants provided informed consent in accordance with the Declaration of Helsinki, and all research projects were approved by the local institutional review boards.

Urbanisation and innovation index

To avoid biases in development assessments caused by reliance on a single indicator, such as the urban population ratio, we integrated approaches of previous research (Shen and Zhou, 2014; Xu et al., 2016; Y. Liu et al., 2017; Bai et al., 2018; Xiong and Xu, 2018; Zhang and Wang, 2018; Lyu et al., 2019; Ping et al., 2020; Cai et al., 2021; Yu, 2021) and a comprehensive understanding of new-type urbanisation to construct an urbanisation index that includes multiple dimensions: population, economy, education and health, urban facilities, and urban environment. Additionally, we developed an innovation index based on two dimensions: innovation inputs and innovation outputs. The specific dimensions are explained as follows:

Population: Population migration from rural to urban areas is the most striking feature and outcome of the urbanisation process. This agglomeration involves not only an increase in the urban population size but also a rise in urban population density. Therefore, we selected three indicators: urban population, the ratio of urban population, and urban population density.

Economy: Urbanisation is both a driving force and a result of economic development. This process is reflected in the overall economic development and the industrial structure of the city, as well as in the economic level of individuals. To capture both dimensions, we selected four indicators: GDP per capita, non-primary industry GDP share, disposable income, and consumption expenditure of urban households per capita.

Education and health: The urbanisation leads to the concentration of the population in cities, requiring more efficient allocation of resources for education and health services. In turn, it enhances education and improves the health of the population, fostering human capital that supports high-quality urban development and innovation. For this dimension, we selected four indicators: educational funds per capita, number of students per 100,000 population in higher education, number of city medical technical personnel in health care institutions per 10,000 population, and beds of urban medical institutions per 10,000 population.

Urban facilities: People-centred urbanisation demands improved infrastructure and public services to enhance cities' carrying capacity and sustainability. To reflect these needs, we selected seven indicators: built-up areas, coverage rate of urban population with access to gas and tap water, public transportation vehicles per 10,000 population, public toilets per 10,000 population, per capita area of paved roads, and public green areas.

Urban environment: A favourable urban living environment is a key factor for sustainable urbanisation and reflects the level of urban environmental governance. It also contributes to the quality of life for urban residents. We selected three indicators in this dimension: green coverage of built-up areas, volume of wastes disposed, and daily disposal capacity of city sewage.

Innovation inputs and outputs: To reflect the elements involved in innovation, we divided this dimension into input and output components. For the input dimension, we selected the indicators of funds and personnel directly involved in innovation: intramural expenditure on R&D funds and full-time equivalents of R&D personnel. For the output dimension, we selected two indicators: domestic patent applications granted and scientific papers taken by a major foreign referencing system. Patents serve as a reliable indicator of innovation activities, while the number of scientific papers published reflects the results of innovation activities.

The specific index system and descriptions of indicators are detailed in the Supplementary Table 1. The urbanisation and innovation indices were calculated using the entropy method (Wang et al., 2015; He et al., 2017) (Supplementary Tables 2 and 3) based on statistical data of indicators. To account for possible lag and cumulative effects of urbanisation, data were collected from 2007 to 2022, and all data were sourced from the

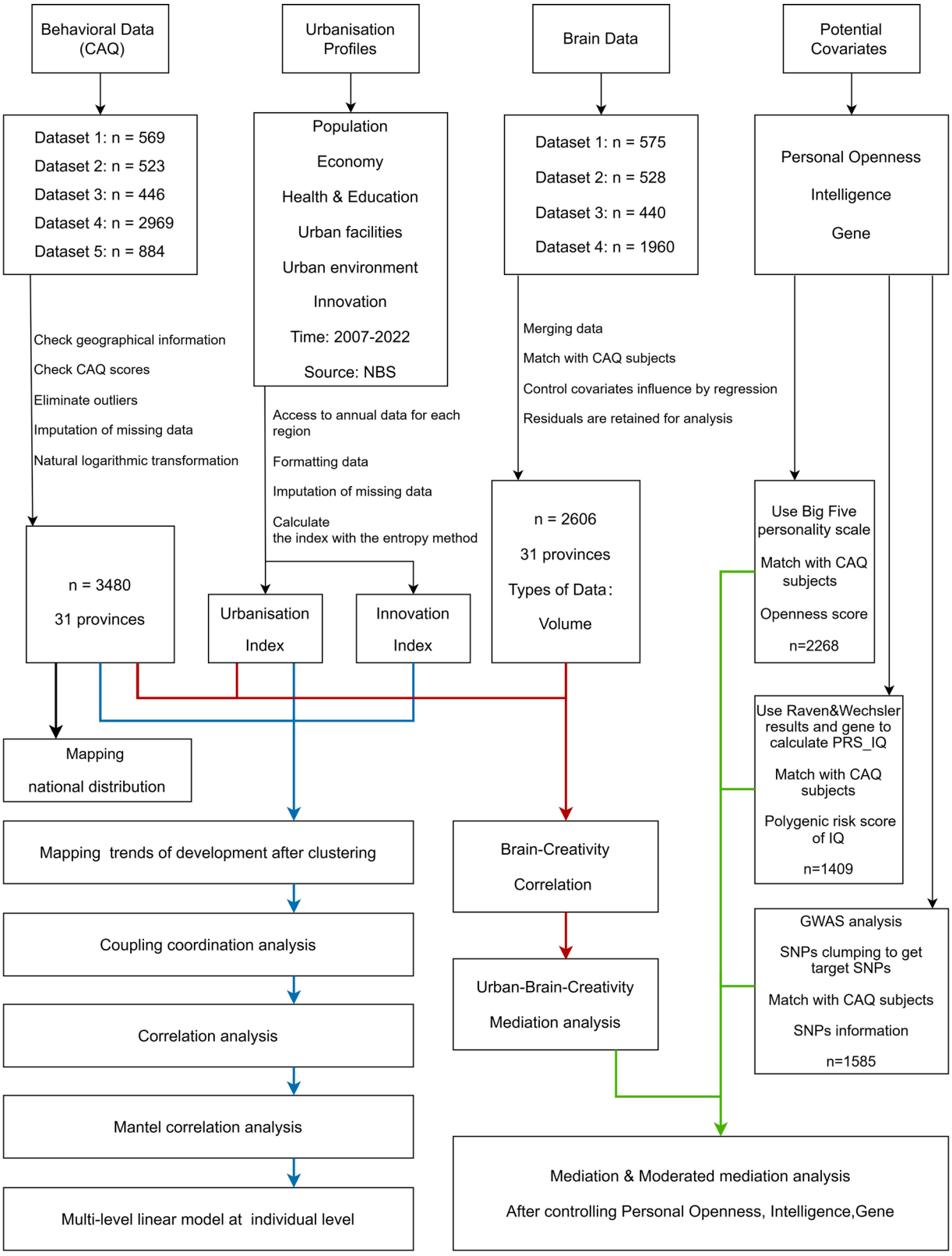


Fig. 1 The flowchart of the study design. CAQ Creative Achievement Questionnaire, GWAS genome-wide association analysis, PRS_IQ polygenic risk score of Intelligence, SNP single-nucleotide polymorphism, NBS National Bureau of Statistics of China.

National Bureau of Statistics of China (NBS) database (https://data.stats.gov.cn/) and *China Science and Technology Statistical Yearbook*. A small amount of missing data was addressed through interpolation based on location. Missing data in the middle were imputed using the mean value of surrounding data, while those at the ends of the range are supplemented with results calculated using the average annual growth rate. The total interpolation ratio is 0.17%.

The procedure and formula for the entropy method are presented in the Supplementary Notes.

Individual creativity measurement

In this study, we operationalise individual creativity as individual creative achievement, defined as externally recognised creative accomplishments/outcomes across domains, rather than internal creative processes (e.g., creative thinking) or dispositional traits. Individual creativity was assessed using the Creative Achievement Questionnaire (CAQ), a widely used self-report measure based on the theory of domain-specific creativity (Carson et al., 2005). While measures of everyday creativity often capture routine problem-solving or latent creative potential, the CAQ specifically assesses publicly recognised, real-world creative achievements, reflecting sustained engagement in creative activities and cross-domain behavioural expressions of creativity. The CAQ is designed to be user-friendly, relatively objective, and resistant to social desirability bias (Silvia et al., 2012), and evaluates creative accomplishments across ten domains, including visual arts, music, dance, architectural design, creative writing, humour, inventions, scientific discovery, theatre, and culinary arts. Each domain (e.g., creative writing) contains eight items in which different achievement statements were described from "no accomplishment" (e.g., "I do not have training or recognised talent in creative writing") to "top accomplishment" (e.g., "my work has been reviewed in national publications"). Some items require participants to indicate the frequency of their accomplishments (e.g., number of publications). A domain-specific CAQ score is calculated by summing the eight items within each domain, and a total CAQ score (domain-general creativity) is obtained by summing across all domains. Since CAQ scores typically exhibit a highly skewed distribution, as only a small number of individuals attain high achievements (Carson et al., 2005), we applied a natural logarithmic transformation to mitigate skewness: $\mathrm{CAQ(ln)} = \ln(\mathrm{CAQ\ scores} + 1)$ (Silvia et al., 2012).

Due to the skewness (skewness = 1.217) of the CAQ scores (Carson et al., 2005; Chen et al., 2014; Zhu et al., 2016; Sunavsky and Poppenk, 2020; Wertz et al., 2020), a log-transformation was applied (Silvia et al., 2012; Nori et al., 2018), and the skewness was improved after this transformation (skewness = −0.327). The log-transformed CAQ scores were used in all subsequent statistical analyses.

Image acquisition and image preprocess

Brain structural data were acquired on a Siemens 3T scanner (Siemens Medical Systems, Erlangen, Germany) at the Brain Imaging Center, Southwest University. High-resolution T1-weighted structure images were obtained using a magnetisation prepared rapid acquisition gradient-echo (MPRAGE) sequence (TR/TE = 1900 ms/2.52 ms, FA = 9 degrees, FOV = 256 × 256 mm^2; slices = 176; thickness = 1.0 mm; voxel size = 1 × 1 × 1 mm^3). All images were processed using fMRIPrep 1.4.1rc1 (RRID:SCR_016216) (Esteban et al., 2019), which is based on Nipype 1.2.0 (RRID:SCR_002502) (Gorgolewski et al., 2011). The fMRIPrep pipeline uses a combination of software packages, including FSL, ANTs, FreeSurfer and AFNI. Each T1-weighted image was manually checked for image quality. Cortical reconstruction and volumetric segmentation were performed using FreeSurfer. The surface-based morphometry (SBM) methodology followed standard procedures in the FreeSurfer (Fischl et al., 2002, 2004; Ségonne et al., 2007). Specifically, this process includes automated Talairach transformation, segmentation of subcortical white matter and deep grey matter volumetric structures (Fischl et al., 2002, 2004), and gyral and sulcal-based parcellation into cortical regions, with extraction of cortical thickness, surface area, and grey matter volume (Fischl et al., 2004; Desikan et al., 2006). Quality control procedures were applied to the segmentation results, and any errors were manually corrected. In this study, we used Desikan–Killiany (Desikan et al., 2006) atlas for segmentation, which includes 34 brain regions in each hemisphere. For further details, refer to the corresponding section on workflows in the documentation for fMRIPrep and FreeSurfer.

Description of development in urbanisation, innovation, and individual creativity

By calculating the mean values of CAQ at the provincial level from 2012 to 2023, we annotated each provincial mean value on a standard map of China, with each province's colour representing the mean CAQ value. Hong Kong, Macao, and Taiwan are excluded.

Clustering analysis of urbanisation level. To classify provinces based on urbanisation levels, we applied *k*-means clustering using the factoextra package in R (Alboukadel Kassambara and Fabian Mundt, 2020). The urbanisation index for each province across years served as the clustering indicator. The ideal number of clusters was determined using the elbow method (Supplementary Fig. 1a), identifying three distinct groups, characterised by higher, medium, and lower urbanisation levels (Supplementary Table 4). The between sum of squares/total sum of squares was 87.30%, indicating a strong separation among the three urbanisation levels (Supplementary Fig. 1b). A classification difference test (Kappa consistency test) was performed to assess the external validity of the clustering. We compared it with the China Central Government's official delineation of the Eastern, Central, Western, and Northeastern regions, which is based primarily on the regional economy and development (National Bureau of Statistics of China, 2011). The higher-level provinces largely overlapped with the Eastern region, the lower-level provinces with the Western region, and the medium-level provinces with the Central region. Considering the difference in the number of categories, we excluded the northeastern provinces for both. The Kappa consistency test suggested a moderate consistency between these two classifications (Kappa = 0.53, $p < 0.001$). When the test only included the provinces with higher and lower urbanisation levels, the Kappa value increased to 0.87 ($p < 0.001$), providing strong support for the reliability and validity of our urbanisation-based regional categorisation.

Trends in urbanisation, innovation, and individual creativity. The urbanisation index and innovation index, covering the period 2007–2022, were used to display trends in urbanisation (Supplementary Fig. 2a) and innovation (Supplementary Fig. 2b). To illustrate the overall trend of individual creativity across different levels of urbanisation, we calculated the mean values of provinces grouped by urbanisation levels at each year. To minimise the discrepancies caused by variations in urban–rural sampling ratios and regional populations, the results were adjusted using weighted calculations, considering the intra-regional urban–rural population ratio and the inter-regional total population ratio.

Coupling coordination analysis. To evaluate how the relationships among urbanisation, innovation, and individual creativity evolve across regions with different urbanisation levels and over time, we employed coupling coordination analysis, a method commonly used in urban development and sustainability research (Lu et al., 2019; Li and Yi, 2020; Tomal, 2021; Liao and Zhang, 2024; Zhang et al., 2024). This approach assesses the dynamic interactions between CAQ, urbanisation index, and innovation index. For the detailed calculation method, refer to the Supplementary Notes. Based on the provincial-level coupling coordination results, termed the *D* value, we calculated the mean values

for regions with different levels of urbanisation and plotted trends in coupling coordination levels over time.

Associations between urbanisation, innovation, and individual creativity

Correlation analysis. We used Pearson's correlation to examine the associations between CAQ, urbanisation index (UR), and innovation index (IN) at the provincial level covering the period from 2012 to 2023. Given that previous studies (Gelfand et al., 2006; Harrington and Gelfand, 2014; Chua et al., 2015, 2019) have identified strong links between cultural tightness, urbanisation, innovation, and creativity, we also examined the correlations between cultural tightness and CAQ, cultural tightness and UR, as well as cultural tightness and IN at the provincial level to validate these findings. As cultural values tend to remain relatively stable over time (Eckhardt, 2002), we sourced data on cultural tightness from the work of Chua et al. (2019).

Mantel analysis. We performed a Mantel analysis using the linkET package implemented in R (Houyun, 2021) to examine the relationships between mean values of urbanisation and innovation sub-indicators and CAQ at the provincial level, as well as between different urbanisation categories and CAQ. Additionally, following the approach used in cultural tightness-related research (Chua et al., 2019), we used urbanisation data from 5 years preceding the CAQ collection period (from 2007 to 2018), and analysed the mean values across these intervals.

Multilevel regression analysis

To further test the hypothesis that urbanisation has a lagged and cumulative effect on individual creativity, we applied a multilevel linear regression model, incorporating individual-level CAQ scores and provincial-level urbanisation and innovation data. The independent variable was the provincial urbanisation index, while the dependent variable was individual-level CAQ, with province included as a random intercept. To account for cumulative and lagged effects, we used the mean urbanisation index from the five years preceding the CAQ collection year, the 5-year prior index, and the current-year index separately. We tested three different models (Supplementary Table 5). Model 1 served as a null model to assess the necessity of using a multilevel model, while Model 2 incorporated age, gender, and data collection time as control variables. Model 3 included age, gender, data collection time, and interaction terms between data collection time and urbanisation index as control variables. Model selection was based on information criteria and parameter comparison. Given its parsimony and clarity, Model 2 was selected for subsequent analyses. Detailed formulas and modelling procedures are provided in the Supplementary Notes. All regression analyses were conducted using R (version 4.3.3).

Relationships between urbanisation, the brain, and individual creativity

Associations between grey matter volume (GMV) and individual creativity. Based on a review of prior research, we selected 26 brain regions of interest (ROI) that were reported to have close associations with individual creativity or CAQ (Supplementary Data 1). Using the lme4 package in R (Douglas Bates et al., 2015), we applied a multilevel regression model with CAQ as the dependent variable and grey matter volume of each ROI as the independent variable, while incorporating age, gender, MRI scanner types, and estimated total intracranial volume (eTIV) as covariates, and provinces as a random intercept, with effect sizes represented by *T*-values. The FDR correction ($p < 0.05$) was applied to identify ROIs significantly correlated with CAQ. Given the limited effect sizes of single brain regions and for better interpretability, we categorised significant ROIs into functional brain networks.

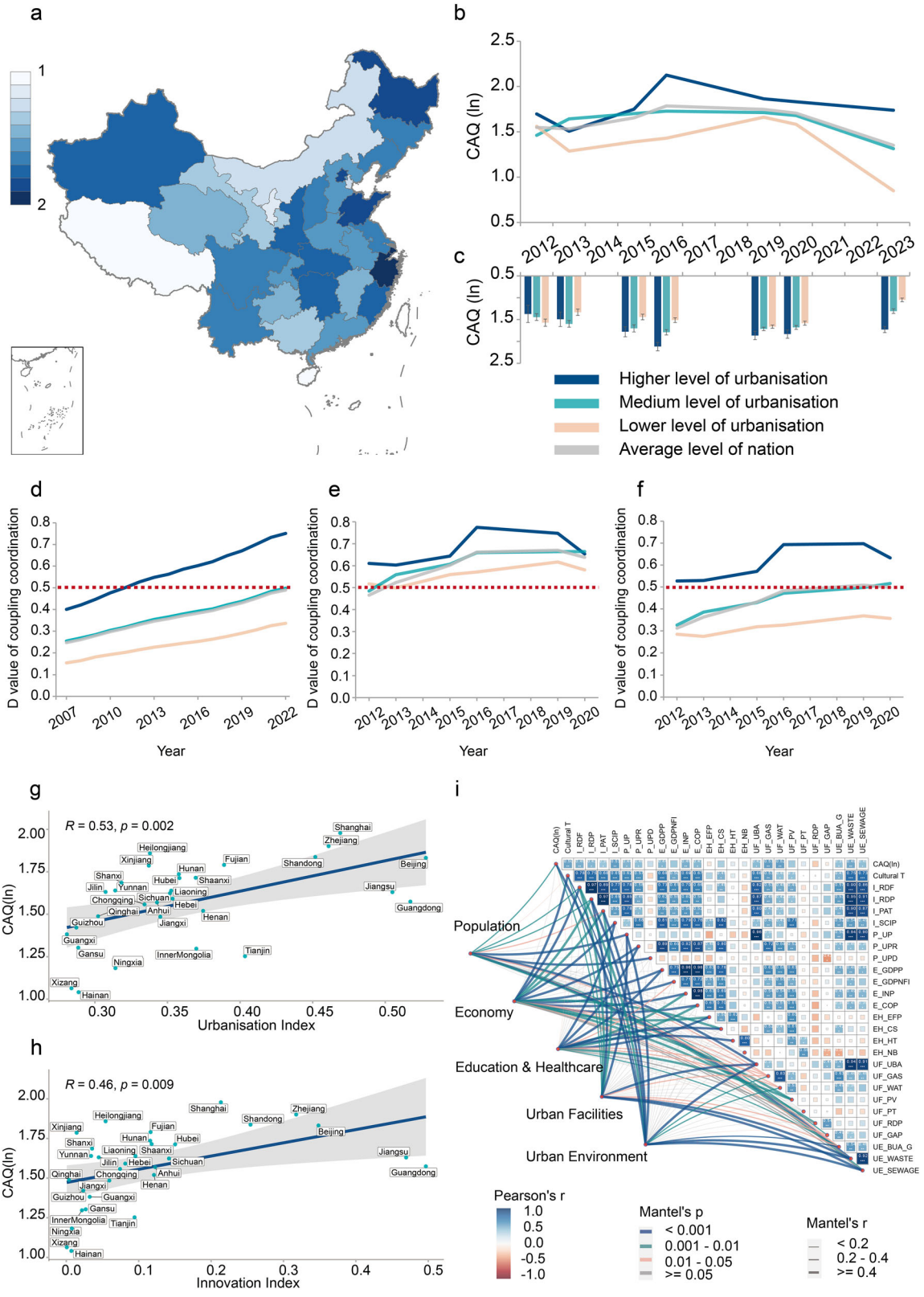
Mediation analysis. To investigate whether brain structural phenotypes mediate the relationship between urbanisation and individual creativity, we conducted a multilevel mediation analysis using the bruceR package in R (Bao, 2024). In the mediation model, individual CAQ serves as the dependent variable, urbanisation index at a time lag of 5-year average, at a time lag of 5 years, or at a time lag of 0 years, as the independent variable, and GMV of functional brain networks showing significant associations with CAQ as mediating variables. Age, gender, MRI scanner types, and estimated total intracranial volume (eTIV) were included as covariates, and province was treated as a clustering variable. Separate multilevel mediation analyses for each functional network were performed, and parameter estimates and 95% confidence intervals were obtained via Monte Carlo sampling. To avoid the basis of selection of ROIs, we performed the same mediation analyses for whole-brain parcellations identified by Desikan-Killiany, comprising 68 parcellations.

Potential covariates

Considering the potential influence of individual personality, intelligence, and genetic factors on individual creativity, we incorporated those covariables into the above mediation models.

Personality openness measurement. Research over decades has consistently shown that personality traits are reliable predictors of both current and future creative performance (Soldz and Vaillant, 1999; Feist and Barron, 2003). Among these traits, openness to experience stands out as the strongest predictor of creativity, such as creative achievement (Silvia et al., 2014; Kaufman et al., 2016; Oleynick et al., 2017; Lebuda et al., 2021). In this study, we assessed individual personality using the Chinese version of the NEO-PI-R, revised by Yang et al. (1999), which has good reliability and validity among the Chinese population. More details on measurements can be found in Supplementary Notes, Dataset information. The personality openness scores were obtained from the participants using the Big Five Personality scale concurrently with the CAQ data collection. By filtering the participant data based on CAQ and regional information, the corresponding openness scores were extracted.

Genetic variation and CAQ: genome-wide association analysis. To explore the association between host genetic variation and CAQ scores, we conducted a genome-wide association study (GWAS) using PLINK, analysing 10 million genetic variants. Following data preprocessing (details in the Supplementary Notes), a linear regression model was applied to examine associations between SNPs and quantitative CAQ scores. Covariates included age, sex, the first 10 principal components (PCA 1–10), and batch effects in the models to control for potential confounding factors. Genetic variants were considered significant if $p < 5 \times 10^{-6}$ (Supplementary Data 10). After that, a clumping procedure was performed to identify independent significant SNPs, selecting the most significant variant per locus, with a threshold of $r^2 > 0.2$ within a 1 Mb window, to avoid the inclusion of highly correlated variants. To incorporate SNP-based individual genetic phenotypes as control variables, we followed the previous research (Zhang et al., 2018) to perform dummy coding for SNP genotypes, that is, genotypes with the reference risk allele (including both homozygous and heterozygous) were coded as 1, while those with the homozygous non-reference allele were coded as 0.



Calculation of polygenic risk score (PRS) for intelligence. Research consistently reports positive correlations between intelligence and creativity (Kim, 2005; Batey and Furnham, 2006; Kim et al., 2010; Karwowski et al., 2021). Thus, we also measured individual intelligence scores using the combined Raven's test (CRT), Wechsler Adult Intelligence Scale, and Raven advanced progressive matrices. More details were available in the Supplementary Notes Dataset information.

Given that different datasets employed varying intelligence assessment methods, we standardised the intelligence scores by computing PRS to eliminate discrepancies arising from methodological differences. The PRS for intelligence was calculated

Fig. 2 Description of the relationship between individual creativity, urbanisation, and innovation. **a** The geospatial distribution of mean scores of Creative Achievement Questionnaire (CAQ, i.e., individual creativity) in each province of China, covering the period from 2012 to 2023. Darker shades of blue represent higher CAQ scores. Trend of individual creativity at different urbanisation levels over the period 2012–2023. Trend with weighted mean (**b**), considering urban and regional population ratio, and the difference histogram of unweighted mean scores across collection year, marked with standard error (**c**). The results of coupling coordinated analyses. Coupling coordinated results of urbanisation and innovation (**d**), CAQ and urbanisation (**e**), CAQ and innovation (**f**). Correlation between creativity, urbanisation, and innovation. The scatter plot for CAQ and urbanisation index (**g**), CAQ and innovation index (**h**), both at the provincial mean level, covering the period from 2012 to 2023 for CAQ, and the period from 2012 to 2022 for urbanisation and innovation index. **i** Correlation between categories and all indicators by Mantel analysis. Cultural T Cultural tightness, I_RDF intramural expenditure on R&D (research and experimental development), I_RDP full-time equivalents of R&D personnel, I_PAT domestic patent applications granted, I_SCIP Chinese scientific papers taken by major foreign referencing system, P_UP urban population, P_UPR ratio of urban population, P_UPD population density of urban area, E_GDPP GDP (gross domestic product) per capita, E_GDPNPI non-primary industry GDP share, E_INP per capita disposable income of urban households, E_COP per capita consumption expenditure of urban households, EH_EFP educational funds per capita, EH_CS number of students per 100,000 population in higher education, EH_HT number of city medical technical personnel in health care institutions per 10,000 population, EH_NB beds of urban medical institutions per 10,000 population, UF_UBA area of built districts, UF_GAS coverage rate of urban population with access to gas, UF_WAT coverage rate of urban population with access to tap water, UF_PV number of public transportation vehicles per 10,000 population, UF_PT number of public lavatories per 10,000 population, UF_RDP per capita area of paved roads, UF_GAP per capita public green areas, UE_BUAG green coverage of built-up areas, UE_WASTE volume of wastes disposed, UE_SEWAGE daily disposal capacity of city sewage, the head of I, P, E, EH, UF, UE refer to the dimension of Innovation, Population, Economics, Education and Healthcare, Urban Facility, and Urban Environments. For Mantel analysis, the calculation was based on the mean CAQ scores of each province from 2012 to 2023 and the average statistical data of each province from 2007 to 2018. All CAQ scores were natural logarithmically transformed.

using PRSice software (Euesden et al., 2015), which integrates summary statistics from the genome-wide association study (GWAS) on intelligence (Savage et al., 2018). The PRS calculation involved combining individual IQ scores with genetic data to generate individual risk scores. A clumping-and-thresholding (C–T) approach was applied to ensure only independent genetic variants were considered. The optimal *p*-value threshold for SNP inclusion was selected to maximise the accuracy of the final risk scores.

Mediation analysis after controlling openness, intelligence, and genetic factors. Considering the potential impacts of individual intelligence, openness to experience, and genetic factors on the CAQ, the three additional variables were separately added as covariates in the mediation model of brain grey matter volume mediating the relationship between urbanisation and individual creativity. The mediation analysis was also conducted using the bruceR package in R. The evaluation included parameter estimation and confidence intervals obtained through Monte Carlo sampling. Since no specific path hypothesis was established, the moderation effects for all paths in the mediation model were assessed. For variables that did not exhibit significant mediation effects, we considered treating them as moderator variables to further investigate their influence on the model.

Results

Trends in individual creativity across regions with different urbanisation levels in China from 2012 to 2023. We first performed a descriptive analysis to examine whether individual creativity varied across regions and years. The geographical mapping of mean individual Creative Achievement Questionnaire (CAQ) scores across provinces is shown in Fig. 2a (detailed descriptive statistics in Supplementary Table 6). There were observed ample differences in CAQ across provinces.

To improve cross-regional comparability, we conducted a clustering analysis based on the annual urbanisation index for each province, classifying all provinces into three categories of urbanisation: higher, medium, and lower levels. The 10-year trend in mean CAQ scores across regions with different urbanisation levels is illustrated in Fig. 2b. We applied weighting adjustments based on the relative proportions of urban and rural populations within each region and the total population across regions to address potential sampling bias due to variations in urban–rural proportions and regional population sizes. The results showed that regions with both higher and lower urbanisation levels exhibited a downward trend in mean CAQ scores in the early years, reaching a nadir in 2013. After that, regions with higher urbanisation levels increased until 2016, then declined until 2019, after which the decline became more moderate. In contrast, regions with lower urbanisation levels experienced a rise in CAQ scores until 2019, followed by a decline that accelerated from 2020 onward. Regions with medium urbanisation level showed a slight increase before 2013, remained stable until a noticeable decline began in 2020. Notably, the higher urbanised regions exhibited a more moderate downward trend after the outbreak of the COVID-19 pandemic compared to the other two groups, suggesting that a higher level of urbanisation may serve as a protective factor for individual creativity during societal adverse events. At the national level, the average CAQ score increased before 2016, remained relatively stable for a period, and began to decline in 2019. Overall, the national trend has remained relatively stable over a long period of time.

We also present the unadjusted mean CAQ scores grouped by urbanisation levels (Fig. 2c), revealing significant differences among regions with different urbanisation levels at the time points of 2013, 2015, 2016, 2020, and 2023 (details in Supplementary Table 7). In both Fig. 2b, c, the ranking of urbanisation levels has remained consistent over time: higher-level regions > medium-level regions > lower-level regions, except for 2012 and 2013.

Evolving coupling coordination among urbanisation, innovation, and individual creativity. To evaluate how the interrelationships among urbanisation, innovation, and individual creativity have evolved across regions with different urbanisation levels, we calculated the degree of coupling coordination, termed the *D* value, pairwise between urbanisation index, innovation index, and CAQ scores over time. The results indicated that, despite a consistent upward trend in *D* values across all three levels of urbanisation, only the higher-level regions have achieved coupling coordination between urbanisation and innovation ($D > 0.5$) since 2011, in accordance with established classification criteria (Supplementary Table 8). While regions with medium urbanisation levels nearly reached basic coordination by 2022, regions with lower urbanisation levels have yet to achieve coupling coordination. Moreover, the gap between higher urbanised regions and the other two categories has widened over time, suggesting that the interplay between urbanisation and

innovation may accelerate coordinated development (Fig. 2d, Supplementary Table 9).

In contrast, the coordination between urbanisation and individual creativity has been consistently sustained since 2013 across all urbanisation levels ($D > 0.5$). Nevertheless, higher urbanisation level regions exhibited markedly stronger coordination ($D > 0.6$) compared to medium and lower levels regions, except in 2020 (Fig. 2e, Supplementary Table 10).

The D value between innovation and individual creativity remained below the threshold ($D < 0.5$) in medium and lower urbanisation regions over time, indicating a persistent lack of coordination until medium-level regions exceeded the threshold in 2020. In contrast, higher urbanisation regions consistently exhibited coordinated development ($D > 0.5$) throughout the study period (Fig. 2f, Supplementary Table 11). At the national level, the average D values across the three index pairs closely resembled those observed in regions with medium urbanisation levels (Fig. 2d–f). Detailed D values for each province were available in the Supplementary Tables 9–11.

Individual creativity associated with urbanisation and innovation. To quantify the relationships between urbanisation, innovation, and individual creativity, we conducted correlation analyses at the provincial level. Mean CAQ scores were significantly positively correlated with both the urbanisation index ($r = 0.53$, $p = 0.002$) and innovation index ($r = 0.46$, $p = 0.009$) (Fig. 2g, h).

Given previous evidence linking cultural factors to urbanisation, innovation, and creativity (Gelfand et al., 2006; Harrington and Gelfand, 2014; Chua et al., 2015, 2019), we also examined the role of cultural tightness. Results showed that cultural tightness was significantly associated with the urbanisation index ($r = 0.77$, $p < 0.001$), the innovation index ($r = 0.76$, $p < 0.001$), and CAQ scores ($r = 0.50$, $p = 0.004$) (Supplementary Fig.3).

We further explored the associations between individual creativity, cultural tightness, and sub-indicators of urbanisation and innovation at the provincial level using Mantel's test. Significant positive correlations were observed between CAQ, cultural tightness, and various sub-indicators, including R&D (research and experimental development) funds and personnel, patents granted, scientific papers, urban population, ratio of urban population, GDP per capita, per capita disposable income and consumption expenditure of urban households, built-up areas, coverage of gas and tap water, green coverage of built-up areas, treatment of waste and sewage (Fig.2i). Moreover, CAQ scores were significantly associated with the four major dimensions of urbanisation: population, education and healthcare, urban facilities, and urban environment (Fig.2i, Supplementary Table 12).

Lagged and cumulative effects of urbanisation on individual creativity. To investigate whether urbanisation has a lagged and cumulative effect on individual creativity, we performed multilevel linear regression analyses to predict individual creativity (i.e., CAQ scores) using the provincial urbanisation index from five years prior, the average over the preceding five-year period, and the current year. As shown in Table 1, all three predictors significantly and positively contributed to individual creativity: the 5 years prior urbanisation index ($t = 5.555$, $p < 0.001$), the previous five-year average ($t = 4.941$, $p < 0.001$), and the current year's index ($t = 2.539$, $p = 0.016$). Prior to this, we have evaluated model performance and validity by comparing alternative multilevel regression specifications, accounting for potential confounders. The selected Model 2 demonstrated improved interpretability and fit.

Applying the same modelling framework, we next examined the relationships between individual creativity and the innovation index, as well as cultural tightness. The 5-year lagged innovation index significantly predicted CAQ scores ($t = 4.562$, $p < 0.001$, Supplementary Table 13), while the current year's index did not ($t = 1.526$, $p = 0.134$, Supplementary Table 13). Cultural tightness also emerged as a significant predictor of CAQ scores ($t = 4.775$, $p < 0.001$, Supplementary Table 14).

We further examined whether specific sub-indicators of urbanisation and innovation could predict individual creativity. Results suggested that multiple sub-indicators from 5 years prior, as well as the 5-year average values, significantly and positively predicted CAQ scores after FDR correction (Fig.3 and Supplementary Table 15). These factors included: total internal expenditure on R&D funds, full-time equivalents of R&D personnel, Chinese scientific papers taken by major foreign referencing system, urban population, ratio of urban population, GDP per capita, disposable income of urban residents per capita, consumption expenditure of urban residents per capita, number of students per 100,000 population in higher education, number of city medical technical personnel in health care institutions per 10,000 population, beds of urban medical institutions per 10,000 population, area of built districts, coverage rate of urban population with access to tap water, number of public transportation vehicles per 10,000 population, per capita area of paved roads, volume of wastes disposed, and daily disposal capacity of city sewage. In contrast, most sub-indicators from the current year could not predict CAQ scores, except for the public transportation vehicles per 10,000 population (Fig. 3 and Supplementary Table 15).

Collectively, these results underscore the lagged and cumulative nature of the effects of urbanisation and innovation on individual creativity. They also highlight the multifaceted economic development, infrastructure, education, health, and environmental components that may contribute to creative outcomes over time.

Structural brain morphology mediated the effects of urbanisation on individual creativity. To investigate whether the brain structural morphology influences the process of urbanisation shaping individual creativity, we conducted mediation analyses using the mean urbanisation index over the preceding five years, brain grey matter volume (GMV), and CAQ scores. The five-year mean index was chosen to better capture potential lagged and cumulative effects of urbanisation exposure on brain morphology and creativity outcome.

We first assessed the associations between CAQ and GMV of 26 candidate creativity-related regions of interest (ROIs, see Supplementary Data 1), while controlling for age, gender, MRI scanner types, and estimated total intracranial volume (eTIV). Significant negative associations were found in the bilateral medial orbitofrontal cortex, left precuneus cortex, left paracentral gyrus, left precentral gyrus, right caudal middle frontal lobe, and right insula (Supplementary Table 16), consistent with prior studies linking brain structural morphometry and creative achievement (Jung et al., 2010; Chen et al., 2014; Zhu et al., 2016; Shi et al., 2017; Wertz et al., 2020). To mitigate potential bias arising from the selection of ROIs, we also performed whole-brain analysis using the Desikan–Killiany atlas, which comprises 68 parcellations (Supplementary Fig. 4, Supplementary Table 17). Except for the left rostral middle frontal gyrus, no other non-ROI regions showed significant associations.

Given the limited effect sizes of individual ROI and the pursuit of better interpretability, we grouped the seven regions into four functional brain networks based on a well-established atlas

Table 1 Multilevel linear regression at the individual level with model 2 (lag 5, 5-year mean and 0 years).

Variables	Model 2: A5Y				Model 2: A5Y_mean				Model 2: A0Y			
	Est (SE)	t	p value	95% CI	Est (SE)	t	p value	95% CI	Est (SE)	t	p value	95% CI
Fixed effects												
Constant	0.751(0.231)	3.251	0.001	0.295,1.208	0.684(0.253)	2.702	0.008	0.182,1.186	1.980(0.326)	6.066	0.000	1.338,2.622
Urbanisation index	2.116(0.381)	5.555	0.000	1.337,2.894	2.074(0.420)	4.941	0.000	1.216,2.933	1.047(0.412)	2.539	0.016	0.206,1.888
Gender: Male	−0.088(0.029)	−3.048	0.002	−0.144, −0.031	−0.086(0.029)	−2.992	0.003	−0.143, −0.030	−0.077(0.032)	−2.418	0.016	−0.140, −0.015
Gender: Female	.	.	.	.	.	.	.	.	.	.	.	.
Age	−0.010(0.009)	−1.163	0.245	−0.027,0.007	−0.010(0.009)	−1.152	0.249	−0.027,0.007	−0.037(0.014)	−2.591	0.010	−0.065, −0.009
Data collection date:2012	0.637(0.102)	6.221	0.000	0.434,0.839	0.636(0.110)	5.804	0.000	0.418,0.853	0.042(0.088)	0.477	0.634	−0.132,0.215
Data collection date:2013	0.573(0.094)	6.119	0.000	0.388,0.759	0.569(0.100)	5.706	0.000	0.371,0.768	0.004(0.079)	0.051	0.959	−0.152,0.160
Data collection date:2015	0.633(0.083)	7.635	0.000	0.469,0.797	0.610(0.085)	7.183	0.000	0.442,0.778	0.085(0.068)	1.261	0.208	−0.048,0.218
Data collection date:2016	0.665(0.071)	9.354	0.000	0.525,0.806	0.654(0.074)	8.841	0.000	0.507,0.800	0.137(0.056)	2.457	0.015	0.027,0.247
Data collection date:2019	0.550(0.053)	10.424	0.000	0.446,0.654	0.545(0.054)	10.040	0.000	0.438,0.652	0.071(0.040)	1.805	0.071	−0.006,0.149
Data collection date:2020	0.465(0.052)	8.931	0.000	0.362,0.567	0.453(0.052)	8.644	0.000	0.350,0.556	.	.	.	.
Data collection date: 2023	.	.	.	.	.	.	.	.	.	.	.	.
Variance components												
Residual	0.619(0.015)	.	0.000	0.591,0.649	0.619(0.015)	.	0.000	0.590,0.649	0.602(0.017)	.	0.000	0.571,0.635
Intercept	0.012(0.006)	.	0.046	0.004,0.032	0.015(0.007)	.	0.030	0.006,0.037	0.009(0.006)	.	0.129	0.002,0.032
I CC	0.029				0.020				0.020			

A5Y: the independent variable is the urbanisation index from 5 years prior; A5Y_mean: the independent variable is the average urbanisation index of the previous 5 years; A0Y: the independent variable is the urbanisation index from the same year as the dependent variable. The dependent variable is CAQ scores with a natural logarithmic transformation. All tests are two-tailed and after multiple comparison correction.

(Uddin et al., 2019; Dragomir and Omurtag, 2020). Specifically, the medial orbitofrontal cortex and precuneus cortex regions were assigned to the default mode network (DMN), the paracentral and precentral gyrus to the somatomotor network (SMN), the caudal middle frontal lobe to the central executive network (CEN), and the insula to the salience network (SN).

We then conducted multilevel mediation analysis to examine whether summed GMVs of regions in specific networks mediated the relationships between the mean urbanisation index over the previous 5 years and CAQ scores. Results suggested significant mediation effects for GMV in the DMN ($\beta_a = -0.315$, $p < 0.001$; $\beta_b = -0.099$, $p < 0.01$; $c' = 0.131$, $p < 0.001$; $ab = 0.031$, 95% CI = [0.012, 0.052], Fig. 4a), SMN ($\beta_a = -0.234$, $p < 0.001$; $\beta_b = -0.065$, $p < 0.01$; $c' = 0.135$, $p < 0.001$; $ab = 0.015$, 95% CI = [0.004, 0.028], Fig. 4b), and CEN ($\beta_a = -0.121$, $p < 0.001$; $\beta_b = -0.051$, $p < 0.05$; $c' = 0.137$, $p < 0.001$; $ab = 0.006$, 95% CI = [0.001, 0.013]), Fig. 4c), explaining 19.10%, 10.12%, and 4.27% of the variance, respectively. However, the mediation effect in the SN was not significant (details in Supplementary Data 2).

Robustness checks using urbanization indices from the current year and 5 years prior yielded similar mediation patterns for the DMN, SMN, and CEN. The SN showed a significant effect only in the analysis using the 5 years prior index (details in Supplementary Data 2). Additionally, we performed supplementary mediation analysis using GMV from significant single brain areas and whole-brain parcellations alongside the 5-year average urbanisation index. Consistent with the ROI-based findings, only those regions that previously showed significant associations with CAQ exhibited significant mediation effects (Supplementary Data 3 and 4). Then we plotted scatter plots between CAQ and volume of significant brain regions using regression residuals after controlling for covariates (Supplementary Fig. 5) to further examine the reliability of the data distribution. These results further corroborate the specificity and robustness of the network-level mediation effects.

Mediation effects of brain morphology after controlling for individual openness, intelligence, and genetic factors. Considering the potential influence of individual openness, intelligence, and genetic factors on individual creativity, we incorporated additional covariates into the mediation models of brain structure morphology that affect the relationships between urbanisation and individual creativity. Specifically, we controlled for the openness scores from the Big Five Personality scale, polygenic scores for intelligence, and single-nucleotide polymorphisms (SNPs) associated with CAQ scores.

After adjusting for openness, the mediation effects of the DMN ($\beta_a = -0.355$, $p < 0.001$; $\beta_b = -0.102$, $p < 0.01$; $c' = 0.121$, $p < 0.001$; $ab = 0.036$, 95% CI = [0.015, 0.058]), SMN ($\beta_a = -0.278$, $p < 0.001$; $\beta_b = -0.067$, $p < 0.01$; $c' = 0.124$, $p < 0.001$; $ab = 0.019$, 95% CI = [0.005, 0.033]), and the CEN ($\beta_a = -0.124$, $p < 0.001$; $\beta_b = -0.050$, $p < 0.05$; $c' = 0.127$, $p < 0.001$; $ab = 0.006$, 95% CI = [0.001, 0.013]) remained significant (Supplementary Data 5). After controlling for polygenic scores of intelligence, significant mediation effects persisted for the DMN ($\beta_a = -0.307$, $p < 0.001$; $\beta_b = -0.086$, $p < 0.05$; $c' = 0.156$, $p < 0.001$; $ab = 0.026$, 95% CI = [0.002, 0.052]) and the CEN ($\beta_a = -0.099$, $p < 0.001$; $\beta_b = -0.065$, $p < 0.05$; $c' = 0.156$, $p < 0.001$; $ab = 0.007$, 95% CI = [0.000, 0.015]), but not for the SMN (Supplementary Data 6). Similarly, after controlling for the SNPs related to CAQ, the mediation effects of DMN and CEN remained robust, except in the model that included the rs532958254_G. However, no significant mediation effects were found in the SMN under these genetic controls (Supplementary Data 7).

Indicators	A5Y	Beta.95CI	P.value	A5YM	Beta.95CI	P.value	A0Y	Beta.95CI	P.value
Innovation									
I_RDF		0.078[0.029,0.127]	0.013		0.070[0.019,0.119]	0.016		0.028[-0.017,0.072]	0.437
I_RDP		0.076[0.026,0.125]	0.016		0.061[0.009,0.112]	0.040		0.037[-0.005,0.078]	0.260
I_PAT		0.026[-0.022,0.074]	0.344		0.008[-0.042,0.058]	0.781		0.014[-0.028,0.055]	0.622
I_SCIP		0.096[0.050,0.143]	0.005		0.092[0.045,0.139]	0.003		0.025[-0.020,0.068]	0.463
Population									
P_UP		0.071[0.010,0.130]	0.046		0.077[0.017,0.137]	0.035		0.052[0.004,0.099]	0.213
P_UPR		0.079[0.011,0.145]	0.046		0.101[0.034,0.168]	0.015		0.036[-0.022,0.091]	0.437
P_UPD		-0.017[-0.074,0.039]	0.620		-0.017[-0.079,0.044]	0.670		-0.022[-0.073,0.028]	0.525
Economy									
E_GDPP		0.112[0.053,0.172]	0.005		0.104[0.045,0.162]	0.005		0.022[-0.029,0.072]	0.525
E_GDPNFI		0.026[-0.040,0.089]	0.465		0.065[0.006,0.122]	0.056		0.010[-0.038,0.057]	0.744
E_INP		0.117[0.038,0.195]	0.016		0.130[0.055,0.204]	0.005		0.028[-0.034,0.089]	0.525
E_COP		0.112[0.033,0.190]	0.018		0.147[0.073,0.223]	0.003		0.045[-0.010,0.100]	0.284
Education & Healthcare									
EH_EFP		-0.023[-0.100,0.055]	0.621		-0.021[-0.096,0.053]	0.670		-0.049[-0.109,0.010]	0.284
EH_CS		0.082[0.024,0.141]	0.018		-0.011[-0.087,0.059]	0.781		-0.017[-0.069,0.034]	0.622
EH_HT		0.112[0.039,0.187]	0.016		0.193[0.116,0.271]	<0.001		0.045[-0.005,0.095]	0.260
EH_NB		0.091[0.029,0.154]	0.016		0.162[0.103,0.221]	<0.001		0.047[0.003,0.091]	0.213
Urban Facilities									
UF_UBA		0.088[0.029,0.146]	0.018		0.084[0.025,0.143]	0.019		0.055[0.009,0.101]	0.213
UF_GAS		0.046[-0.011,0.101]	0.146		0.059[0.001,0.115]	0.067		0.023[-0.031,0.074]	0.525
UF_WAT		0.092[0.040,0.143]	0.005		0.062[0.001,0.124]	0.067		-0.010[-0.061,0.043]	0.744
UF_PV		0.037[-0.014,0.089]	0.187		0.086[0.036,0.136]	0.005		0.084[0.045,0.122]	0.003
UF_PT		-0.011[-0.056,0.035]	0.668		-0.041[-0.090,0.009]	0.144		0.026[-0.016,0.068]	0.437
UF_RDP		0.004[-0.061,0.069]	0.914		-0.098[-0.171,-0.028]	0.017		-0.007[-0.064,0.049]	0.826
UF_GAP		0.060[0.003,0.116]	0.059		-0.002[-0.065,0.061]	0.950		-0.028[-0.078,0.022]	0.463
Urban Environment									
UE_BUA_G		0.047[-0.005,0.099]	0.117		-0.019[-0.084,0.044]	0.664		0.000[-0.047,0.046]	0.997
UE_WASTE		0.072[0.019,0.123]	0.018		0.084[0.032,0.137]	0.010		0.054[0.010,0.097]	0.213
UE_SEWAGE		0.067[0.012,0.121]	0.041		0.056[0.000,0.112]	0.073		0.041[-0.003,0.084]	0.260

-0.1 0.1 -0.1 0.1 -0.1 0.1

Fig. 3 Forest plot of multilevel linear regression using model2 (lag 5, 5-year mean and 0 years). The independents are sub-indicators of urbanisation index and innovation index. The dependent variable is CAQ scores with natural logarithmic transformation. A5Y: the independent variable is the sub-indicators from 5 years prior; A5YM: the independent variable is the average sub-indicators of the previous 5 years; A0Y: the independent variable is the sub-indicators from the same year as the dependent variable. All *p*-values are FDR corrected.

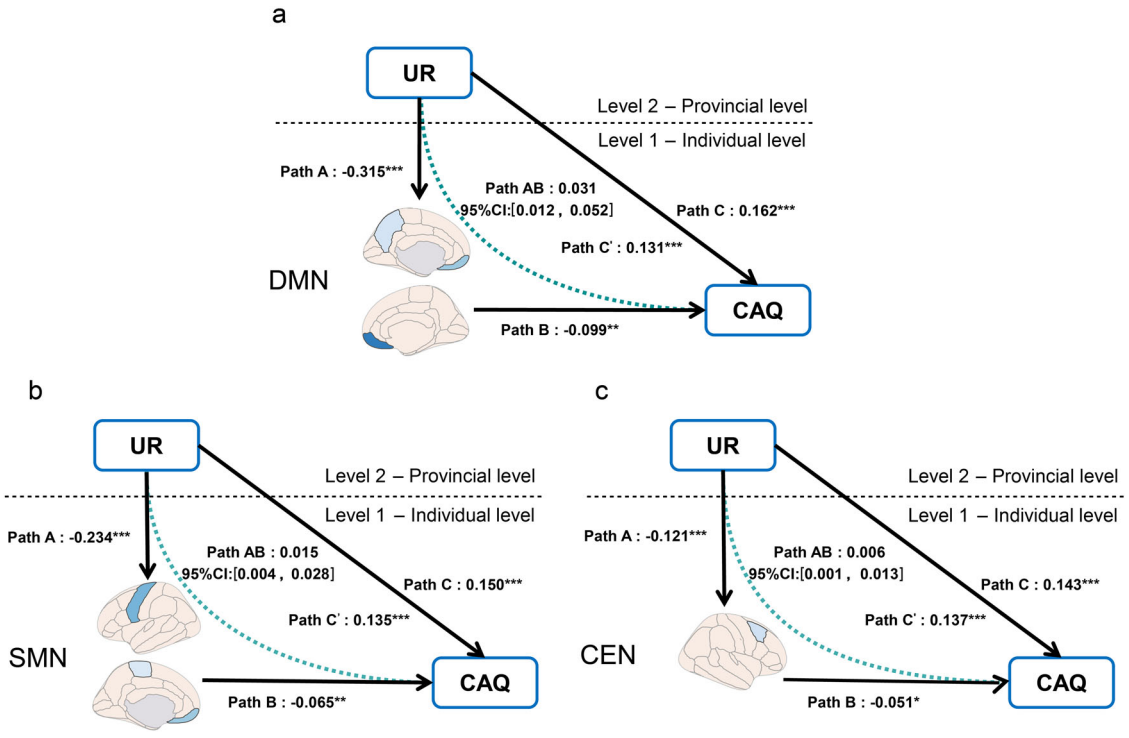


Fig. 4 Results of multilevel mediation analysis between urbanisation, creativity, and brain. Results of mediation between urbanisation index, CAQ and significant regions in DMN (**a**), SMN (**b**), and CEN (**c**). The independent variable is the mean urbanisation index over the previous 5 years, while the dependent variable is CAQ scores with natural logarithmic transformation. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

We also explored whether the unmediated intellectual and genetic variables moderate the mediation models of brain structures' morphology that affect the relationships between urbanisation and individual creativity. Moderation analyses further indicated that neither polygenic scores nor individual SNPs moderated the mediation effects, except for rs532958254_G, which significantly moderated the mediation pathway through GMV in the DMN (Supplementary Data 8 and 9).

In summary, these findings demonstrate that the mediation effects of brain structural morphology, particularly within the DMN and CEN, on the relationship between urbanisation and individual creativity are robust and remain significant after controlling for individual differences in openness, intelligence, and genetic factors. They support the hypothesis that brain plasticity is a key neurobiological mechanism through which urbanisation shapes individual creativity.

Discussion

During new-type urbanisation, we observed not only the coordinated development of urbanisation, innovation and individual creativity in China, but also the emergence of various urban elements that create conditions conducive to fostering creativity. By examining the links between urbanisation, brain structure, and individual creativity, we found that the urbanisation index positively predicted individual creative achievement (CAQ), with brain volume serving as a significant mediator. This effect persisted after controlling for personal openness, intelligence, and genetic factors. These findings suggest that urbanisation, as a foundational environmental condition, may be associated with individual creative achievement and related brain structural differences.

From 1978 to 2012, China experienced a dramatic surge in urbanisation, with the proportion of the urban population rising from under 20% to 52.6%, adding over 500 million urban residents (The Research Center of Development of the State Council and the World Bank Group et al., 2014). However, the rapid growth brought significant challenges such as pollution, overcrowding, and rural labour shortages (Bloom et al., 2008; Yang, 2013). Since the 18th Communist Party of China (CPC) National Congress, the government has prioritised sustainable, people-centred urbanisation, focusing on improving welfare and well-being, aiming to balance development and rectify the legacy issues of rapid urbanisation over the past decades (Pan, 2012; Bai et al., 2014). To capture this transition, we employed a multi-dimensional urbanisation index that moves beyond single-indicator metrics, such as the urban population ratio (Rodrigues and Franco, 2020; Yu, 2021). Our findings reveal a consistent upward trend in the urbanisation index across various regions (Supplementary Fig. 2a), objectively reflecting the progress of the new-type urbanisation development. Concurrently, the whole development process is driven by innovation, which has always been a key focus in China's development agenda. Our results show that, despite regional disparities, innovation levels have steadily increased nationwide over the past decade (Supplementary Fig. 2b). Notably, regions with higher level of urbanisation exhibited a faster growth in innovation, supporting the notion that urban environments provide fertile ground for innovation (Pred, 1966; Higgs, 1971; Zhixian, 2008; Kaiming, 2010; J. Liu et al., 2017).

Sustainable urbanisation requires coordinated development of all parts. Through the coupling and coordination analysis between urbanisation and innovation, we observed positive results; however, consistent with previous research (J. Liu et al., 2017; Duan et al., 2022), we also noted the lag and disconnection caused by rapid development. Except for regions with higher levels of urbanisation, other areas have yet to achieve coupling and coordination, failing to effectively transformed the advantages of urbanisation and innovation into mutual benefits. Nevertheless, the sustained upward trend is encouraging, and the gap across regions can be narrowed through ongoing efforts to strengthen regional coordinated development.

Although empirical evidence is still limited (Štreimikienė and Kačerauskas, 2020), increasing studies have begun to link creativity with sustainability across multiple levels, including urban, social, and individual dimensions (Song and Yu, 2018; Sovacool et al., 2018; Cheng, 2019; Brem and Puente-Díaz, 2020; Hensley, 2020; Kagan et al., 2020; Mróz and Ocetkiewicz, 2021; Ogbeibu et al., 2021; Muisyo et al., 2022; Souto, 2022), and viewed creativity as out-of-the-box solutions that can drive the transition toward a sustainable future (Przychodzen et al., 2016; Awan et al., 2019; Betlej and Kačerauskas, 2021). Consistent with the above opinions, our results on coupled and coordinated development among individual creativity, urbanisation and innovation also exhibit sustainable prospects. Despite regional differences, individual creativity and urbanisation show a positive level of coupling coordination. Individual creativity and innovation have reached basic coupling coordination, except for regions with a lower level of urbanisation, which are progressing toward coordination. Notably, our dataset encompasses samples from both the pre- and post- COVID-19 periods. In the trend of CAQ changes across urbanisation levels, regions with higher levels of urbanisation showed a more gradual decrease compared to the sharp decline observed in other regions, suggesting that urbanisation may offer a protective effect on individual creativity during severe public health crises, contributing to enhanced resilience and individual sustainability. Such differences may partly stem from inequalities in resources and technology (Goudeau et al., 2021).

As a key to integrating sustainable cycles, through correlation and multi-level regression analyses, we found that urbanisation has a significant positive relationship with individual creativity, and a lagged and cumulative effect on predicting individual creativity. Sternberg highlighted that while individuals may have creative potential, it can be stifled without an environment that tolerates reasonable risk or provides at least minimal support for creativity (Sternberg and Lubart, 1991). Urbanisation contributes to this by ensuring a stable and diverse food supply, organised public services, a well-educated population, and the presence of hubs for learning and innovation, while also improving healthcare and reducing child mortality (Dye, 2008). More specifically, through Mantel analysis, we observed that individual creativity is significantly correlated with categories such as population, education and health, urban facilities, and urban environment. Similarly, sub-urbanisation indicators were also found to significantly predict individual creativity, supporting the interpretation that urbanisation is positively associated with individual creativity. In addition to its direct association, urbanisation also indirectly associated with individual creativity through innovation-related indicators. The Mantel analysis revealed that sub-urbanisation indicators and dimensions are significantly correlated with both innovation inputs and outputs, while the atmosphere of innovation formed by these inputs and outputs further influences individual creativity. Our results indicated that the innovation index, innovation inputs (i.e., research funds and personnel) and outputs (i.e., patent grants and scientific papers) are significantly correlated with individual creativity. Moreover, the provincial-level innovation index, research funds and personnel, and the number of cited scientific papers can significantly predict individual creativity.

Taken together, these findings situate individual creative achievement within broader discussions of urban creativity

research and policy. Our results indicate that the relationship between urbanisation and CAQ-based creative achievement is characterised by an overall positive association coupled with regional heterogeneity, reflecting a complex reality where urbanisation expands opportunities for knowledge exposure and innovation participation, yet simultaneously generates disparities in translating creative potential into realised outcomes. This aligns with urban creativity research and policy discussions on the dual and context-dependent nature of "creative urbanisation", highlighting an inherent tension in which urban environments may facilitate access to educational, cultural, and innovation-related resources, while structural disparities in urban development can produce profoundly unequal opportunities for individuals to transform their creative potential into socially recognised achievements. On this basis, further empirical work is needed to unpack the specific micro-level mechanisms through which urbanisation exerts these divergent influences on individual creativity.

The integration of urban economic theories with neuroscientific evidence provides an integrative conceptual framework for understanding how urbanisation reshapes human brain development. Urban land-use theory emphasises how accessibility and land-market mechanisms generate spatial concentration and functional specialisation in cities (Fujita, 2013). Building on this spatial concentration, urban development is shaped by a fundamental trade-off between agglomeration economies and agglomeration diseconomies: spatial proximity reduces interaction costs and facilitates learning and knowledge diffusion, while density simultaneously generates negative externalities such as congestion, rising living costs, and intensified competitive pressures (McCann, 2001; O'Sullivan, 2003; Capello, 2015). This trade-off is particularly relevant to creative processes and regional development, which rely on urban ecosystems, institutions, and networks, while the density and competition inherent in these environments drive the dynamism of creative activity (Dellisanti, 2023). Translating these macro-level principles to individual development, agglomeration benefits can be seen as environmental enrichment, where rich and complex socio-economic inputs support creative processes and higher-order cognition. Consistent with this perspective, the brain regions we identified (e.g., dorsolateral prefrontal cortex, medial orbitofrontal cortex, precuneus, supplementary motor areas, and precentral gyrus) have been implicated in creativity-related and higher-order cognitive functions and may benefit from the rich sensory and social stimulation associated with urban life.

At the same time, urban economics also highlights agglomeration costs—the negative externalities of density—which may impose a "structural toll" through high living costs, intense labour market competition, and elevated social stressors (McCann, 2001; Fujita, 2013; Capello, 2015). Such pressures may help explain observed neuroanatomical alterations associated with urban exposure. For example, Haddad et al. (2015) found that urban exposure is associated with reduced grey matter volume in the prefrontal cortex. Urban upbringing is linked to thinner dlPFC and mPFC in adults (Besteher et al., 2017; Lammeyer et al., 2019; Zhang et al., 2021). Reduced medial prefrontal cortex volume mediates the association between urban living and depressive symptoms (Xu et al., 2022), while the precentral gyrus and right orbitofrontal cortex are linked to mood instability, and the left supplementary motor area to anxiety symptoms (Xu et al., 2023). Together, these findings converge with our results and suggest that brain regions involved in higher-order cognition may be shaped by urbanisation in complex ways, simultaneously supporting adaptive cognitive functions while increasing vulnerability to stress-related outcomes. This neural complexity reflects an adaptation to the "urban paradox"—an environment that serves as a hub of creative opportunity yet imposes chronic socio-economic pressure—highlighting the need for an integrative perspective on the multifaceted neural responses to urban life.

In addition to the impact of the urban environment features on creativity, cultural tightness is an alternative explanation. Cultural tightness refers to the degree to which a society is characterised by rules and norms, and the extent to which people are punished or sanctioned when they deviate from these rules and norms (Gelfand et al., 2006; Chua et al., 2019). Interestingly, culture tightness is positively correlated with levels of urbanisation in China (Chua et al., 2019), whereas in the United States, the relationship is reversed (Harrington and Gelfand, 2014). Our correlation analyses between cultural tightness, urbanisation, innovation, and individual creativity revealed significant relationships among these factors, which are consistent with previous findings (Chua et al., 2019). Moreover, the multilevel regression analysis showed that cultural tightness significantly predicts individual creativity. Building on insights from previous studies (Bloom et al., 2008; Glaeser, 2011), Alfred Marshall proposed that cities accelerate the flow of knowledge and new ideas. However, urbanisation also presents challenges for governance that many developing countries struggle to address effectively; as such, urban residents require more efficient governance structures to function optimally. We propose that tighter cultures, which have stronger social norms, contribute to higher administrative and managerial efficiency in certain regions. This, in turn, maximises the benefits of urbanisation—such as agglomeration and exchange effects—facilitating the incubation of new ideas and technologies. Consequently, this process enhances social returns by increasing human capital, resulting in efficient growth (Black and Henderson, 1999; Montgomery et al., 2003).

Linking to the point above, our results in brain volume support the culture tightness explanation. On one hand, when strong social norms are established (Fehr and Fischbacher, 2003), interdependent orientations (Sober and Wilson, 1999), which are linked to a strong value placed on social obligations and duties (Kitayama et al., 2017), also develop. Cross-cultural studies further indicate that compared to Westerners, Asians exhibit greater interdependence (Nisbett et al., 2001). More importantly, this phenomenon has been identified as being associated with reductions in orbitofrontal cortex (OFC) volume (Kitayama et al., 2017; Wang et al., 2017)—a finding echoed in our study. On the other hand, the social norms may demand strong behavioural regulation and inhibition (Gelfand et al., 2006; Baumeister et al., 2007; DeBono et al., 2011; Gelfand et al., 2011). Consistent with our findings, research on behavioural inhibition sensitivity (Fuentes et al., 2012) has also reported volume reductions in bilateral OFC and the left precuneus. Behavioural inhibition system (BIS) activity leads to sustained behavioural suppression, heightened attention, and enhanced information processing, which are positively correlated with most measures of creativity (Benedek et al., 2012).

Regarding the brain regions identified in the present study, mediation analysis suggests that specific brain regions may mediate the relationship between urbanisation and individual creativity. This finding provides a possible and positive underlying mechanism from a neuroscientific perspective. Research in this field has typically focused on identifying the factors and neural mechanisms associated with adversity, with less attention given to beneficial experiences that may promote resilience and coping abilities of human brain adaptation or buffering against the negative effects of environments (Tost et al., 2015). In contrast, our study highlights the potentially beneficial associations of urbanisation on individual creativity—an important trait that supports personal development and adaptive functioning. Although this positive perspective has been relatively

underexplored, previous research has reported positive associations between urban living and cognitive development. For instance, childhood urbanicity has been linked to enhanced cognitive function (Xu et al., 2022; X. Zhang et al., 2023) and increased GMV in the dorsolateral prefrontal cortex (X. Zhang et al., 2023). As more people live in urban areas, cities provide better educational, economic, and healthcare resources, which may, in turn, provide opportunities that benefit residents, although such benefits are likely uneven and context-dependent. It is crucial to acknowledge these benefits and carefully examine their implications.

Limitation and outlook. Several limitations of this study should be acknowledged. First, as a study linking macro-environmental factors with micro-individual behavioural performance, our findings are inherently influenced by specific temporal, spatial, and sociocultural contexts. These factors may impact the results and constrain the generalisability of the research. Nevertheless, we have endeavoured to mitigate these limitations by using long-term data, constructing multidimensional indicators, analysing multilevel variables, and controlling for potential confounding factors, to faithfully reflect the associations between urbanisation and individual creativity during this specific developmental phase in China. Second, the observed lagged effects of urbanisation index on individual creativity should be interpreted with caution. Although there is a time gap, it remains difficult to draw definitive conclusions regarding causal relationships. Moreover, the mediation results should be regarded as statistical indirect associations rather than evidence that urbanisation causes changes in brain structure, because unmeasured factors such as migration history, family socioeconomic background, educational opportunity, and residential self-selection may also contribute to the observed associations. Third, our sample size does not match that of large-scale datasets typically used in studies focusing on social and economic contexts. However, it is substantial within the specialised domain of individual creativity, which holds a unique value and is not easily replicated. Notably, all measurements, including individual creativity, regional context, brain structure MRI, personality, intelligence, and genetic factors, were collected from the same participants. This integrated, multimodal dataset enhances the internal validity and reliability of our findings. In addition, we employed a well-established instrument to directly assess individual creativity, instead of approaches that rely on indirect or alternative measures. However, the CAQ primarily reflects self-reported creative achievements rather than creative processes or potential. Using the total score may also mask heterogeneity in achievement profiles (e.g., specialisation in one domain vs. breadth across domains). Moreover, CAQ scores can be shaped by opportunity structures that differ by place of residence, as some creative activities and platforms for training, participation, and recognition are more available in larger cities. Thus, the observed urbanisation–CAQ link may partly reflect differential opportunities to realise creative achievements. Furthermore, as a self-report instrument, the CAQ may be subject to biases related to memory accuracy and social desirability. These issues are particularly relevant in the Chinese cultural context, where norms of modesty and self-effacement may influence individuals' willingness to report or emphasise personal achievements, and where self-presentation styles may vary across regions and cohorts. Future work could employ multidimensional and objectively assessed tools to measure creativity. Finally, regarding the granularity of the analyses, we used a provincial-level urbanisation index. We acknowledge that, given China's vast territory and uneven population distribution, this scale may not be sufficiently fine-grained and may obscure within-province heterogeneity. This choice reflects a pragmatic balance among analytical detail, data availability, measurement reliability, and statistical power, enabling us to maximise the value of this unique multimodal dataset. Future work could replicate our framework using finer spatial units where consistently available, and explicitly model within-province heterogeneity.

In addition to the above limitations, there are still some issues that need to be answered in future work.

1. Further validation is needed in diverse cultural and regional contexts to test the generalisability of the relationships between urbanisation and individual creativity. Longitudinal follow-up measurements for individual creativity could also be helpful for identifying critical periods of its growth and transformation.
2. The portrayal and measurement of urbanisation need to be updated. With the advent of the artificial intelligence era and the introduction of generative AI into various dimensions of urban life, human–environment interactions are continuously evolving. This requires us to integrate more advanced technological tools to utilise geospatial information, social networks, and urban landscape elements, thereby improving the temporal and spatial resolution of urbanisation characterisation. It is more urgent to redefine urbanisation in this new wave, capturing the imprints of urbanisation that represent this era, thus better tracking the changing relationship between urbanisation and individual creativity.
3. Changes in brain structure may represent the trade-offs of multiple functions influenced by the external environment. Future work should aim to isolate the individual impacts on various functions. Interdisciplinary collaboration, particularly multi-omics work, may help explore finer-grained associations between urbanisation and human creativity, and clarify the boundary conditions of brain-environment relationships.

In conclusion, our study documented sustained growth in urbanisation and innovation across China from 2012 to 2023, alongside a coordinated development among urbanisation, innovation, and individual creativity. We found significant positive associations among these three factors, and further observed that variations in regional brain volume within the default mode, somatomotor, and central executive networks statistically mediate the association between urbanisation and individual creativity. These findings are consistent with a possible neurobiological pathway linking urbanisation and creativity, but they should be viewed as preliminary evidence of brain plasticity-related associations rather than definitive proof of mechanism. We cautiously propose that new-type urbanisation stimulates innovation, which in turn drives sustainable urban development. Within such environments, individuals may benefit from enriched infrastructure and social resources that support creative achievement, which may ultimately serve as a vital human resource. In this way, individual creativity may emerge not only as a product of urban transformation but also as a contributor to broader societal innovation and sustainability. This kind of dynamic highlights the organic integration of macro-level urban sustainability with micro-level individual sustainability.

Data availability

The processed data generated in this study are available on the Open Science Framework [https://osf.io/qhb4e/?view_only=8388f5a311d54d5c9938013858df2df0]. Some of the raw data used in the present study have been publicly available and analysed in previous studies, and a part of the non-public data is available from

the corresponding author upon reasonable request. The raw sequence data reported in this paper have been deposited in the Genome Sequence Archive (Genomics, Proteomics & Bioinformatics 2021) in National Genomics Data Center (Nucleic Acids Res 2024), China National Center for Bioinformation/Beijing Institute of Genomics, Chinese Academy of Sciences (GSA-Human: HRA009175) that are publicly accessible at https://ngdc.cncb.ac.cn/gsa-human. The release of the data was approved by the Ministry of Science and Technology of China (Project ID: 2025BAT00486). Access to individual-level data must be approved by corresponding authors and is subject to the policies and approvals from the Human Genetic Resource Administration, Ministry of Science and Technology of the People's Republic of China.

Code availability

The code generated for this study has been shared on the Open Science Framework at [https://osf.io/qhb4e/?view_only=8388f5a311d54d5c9938013858df2df0].

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Author contributions

QC, YS, JQ, and XW conceptualised the study; QC, YS, QH, XL, TF, HC, and JQ contributed the data; YS, WY, YZ, and JY performed the analyses; YS and QC wrote the original draft; QC, YS, WY, YX, and XW reviewed and edited. All co-authors provided feedback and approved the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

All procedures performed in studies involving human participants were in accordance with the ethical standards of the Institutional and/or National Research Committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The research projects were approved by the Scientific Review Committee of the Department of Psychology at Southwest University on 3 June 2012 (Approval No. spy-2012-012) and by the Ethics Committee of Chongqing Ninth People's Hospital on 31 December 2019 (Approval No. CNPH2019016).

Informed consent

All participants provided informed consent in accordance with the Declaration of Helsinki. Written consent was obtained by the authors from June 2012 to November 2016 and from December 2019 to December 2020 for participants recruited between 2012 and 2020. For participants recruited in 2023, electronic consent was collected via the Wenjuanxing platform from September to November 2023. The survey platform presented the informed consent statement to every participant before the questionnaire began, and participants were required to read and agree to it before proceeding to complete and submit the questionnaire. The system was configured not to collect any participant information prior to this. Prior to participation, all participants were informed of the study objectives, the confidentiality of their responses, and their right to withdraw at any time without repercussions. All data were anonymized to protect the identities of participants and ensure confidentiality.

Additional information

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